

Machine Learning Basics

Prof. Artur Jordão

The Artificial Intelligence Landscape

Definitions

The Artificial Intelligence Landscape

- Artificial Intelligence
 - Any technique that enables computers to mimic human behavior
- Machine Learning
 - Ability to learn without explicitly being programmed
- Deep Learning
 - Extract patterns from data using neural networks

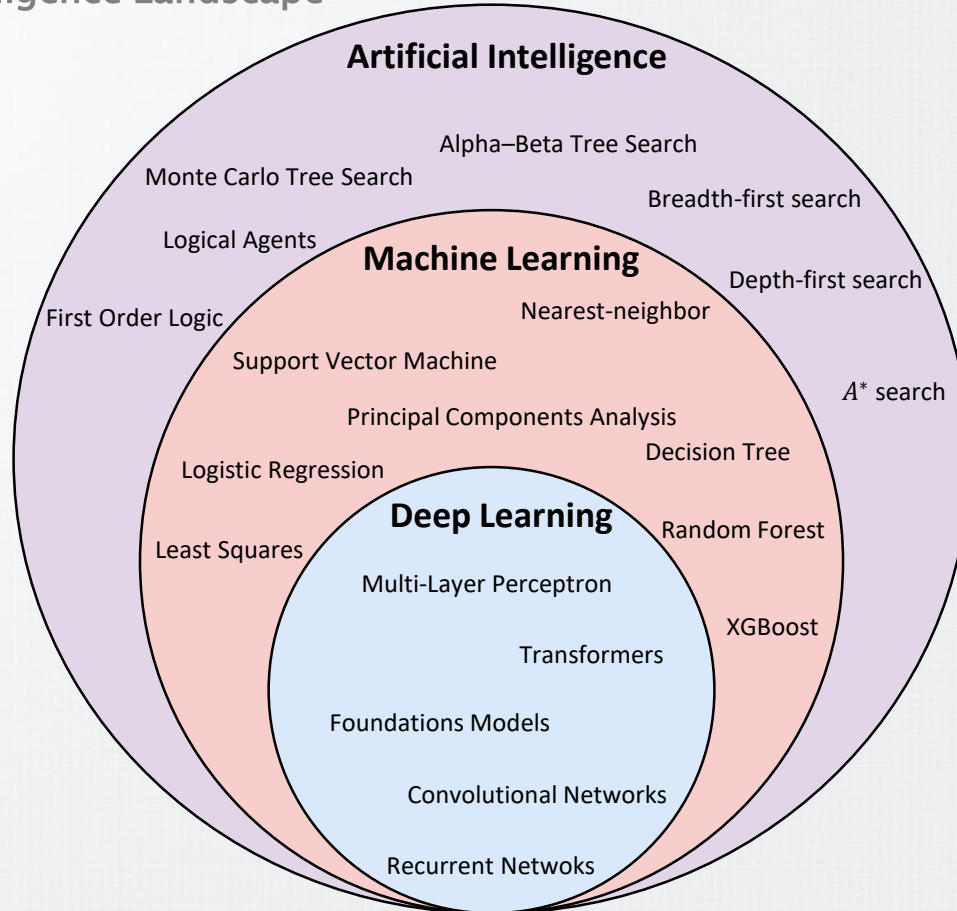
Definitions

The Artificial Intelligence Landscape

- UNESCO
 - *“Artificial intelligence refers to the capability of machines to perform many tasks that typically require human intelligence, such as learning, reasoning, and problem-solving.”*
- REGULATION (EU) 2024/1689 OF THE EUROPEAN PARLIAMENT
 - *“ ‘AI system’ means a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments”*
- Plano Brasileiro de Inteligência Artificial
 - *“[...] conjunto de modelos, algoritmos, técnicas e metodologias que podem ser implementados como sistemas computacionais que produzem resultados como previsões, classificações, recomendações e decisões, a partir de processos de aprendizagem baseados em grande volume de dados, com potencial para influenciar ambientes físicos e virtuais.”*

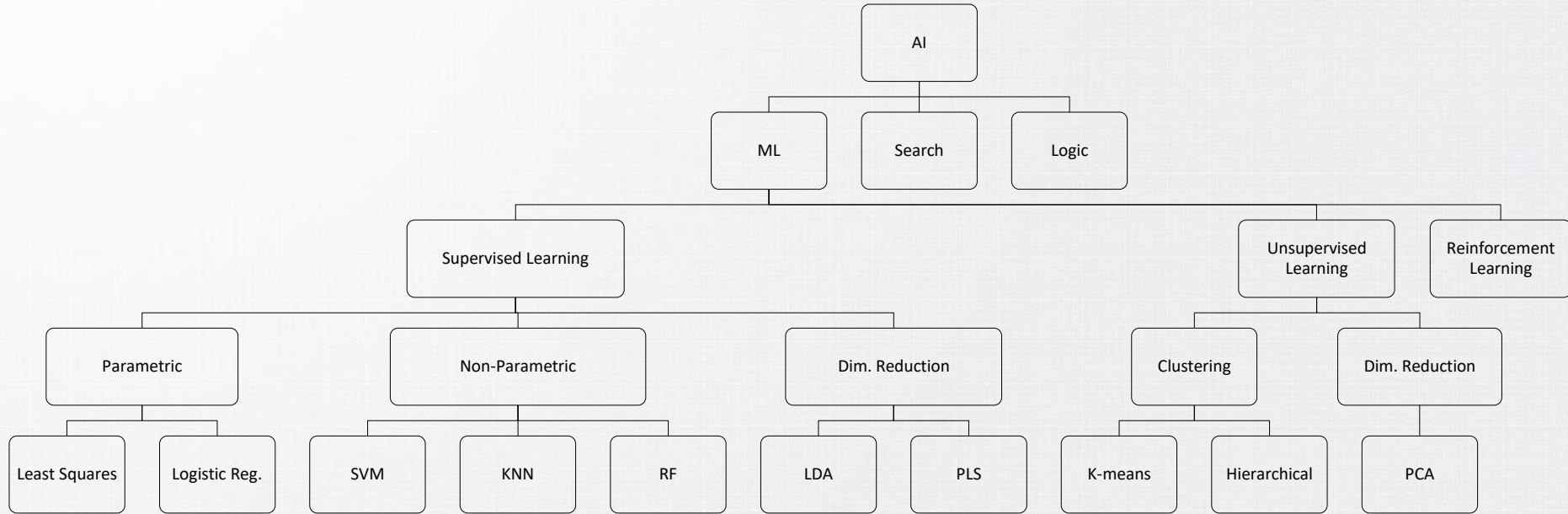
Machine Learning and Deep Learning

The Artificial Intelligence Landscape



Machine Learning and Deep Learning

The Artificial Intelligence Landscape

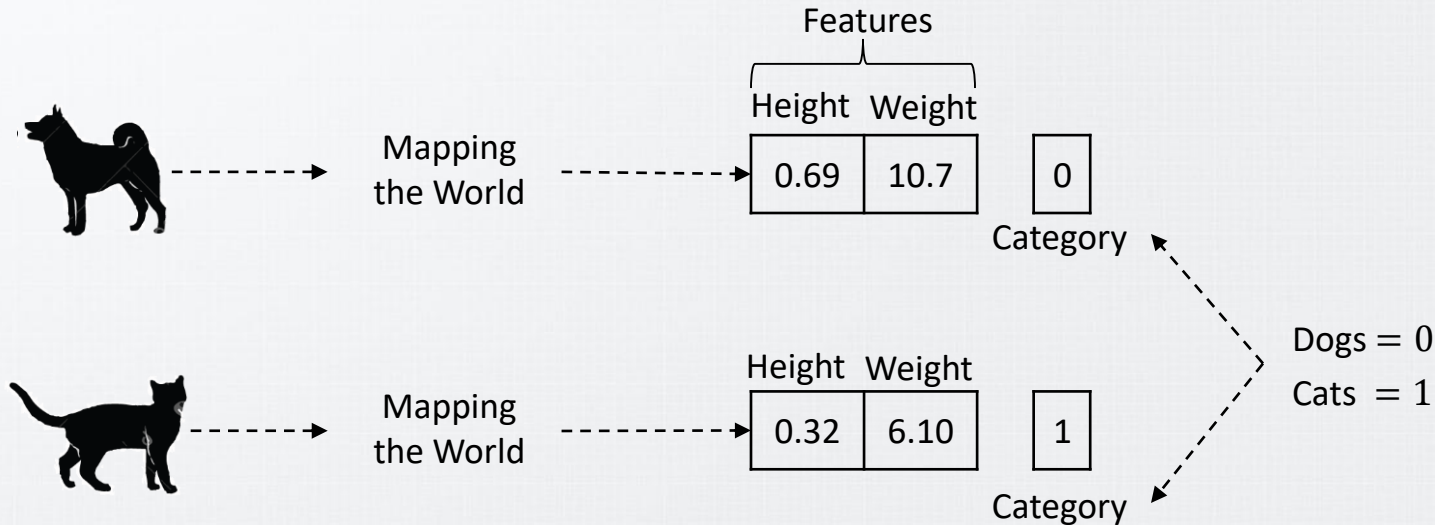


Independent (X) and Dependent (Y) Variables

Preliminaries

Independent and Dependent Variables

- Mapping the state of the world (features – m)
 - How can we represent the states (i.e., objects) of the world?
 - Numerical representation (features engineering)



Preliminaries

Independent and Dependent Variables

- Assume that we represent (map) n dogs and cats using m features
- Let $X \in \mathbb{R}^{n \times m}$
 - Independent variables
 - Data samples – each dog/cat composes a row in X
- Let $Y \in \mathbb{R}^{n \times c}$
 - c stands for the number of categories (i.e., 2)
 - Dependent variable
 - Labels/Classes

	X		Y
n	0.69	10.7	0
	0.85	14.5	0
	0.67	20.3	0
			.
	0.32	6.1	1
	0.21	4.0	1
	m		

Preliminaries

Independent and Dependent Variables

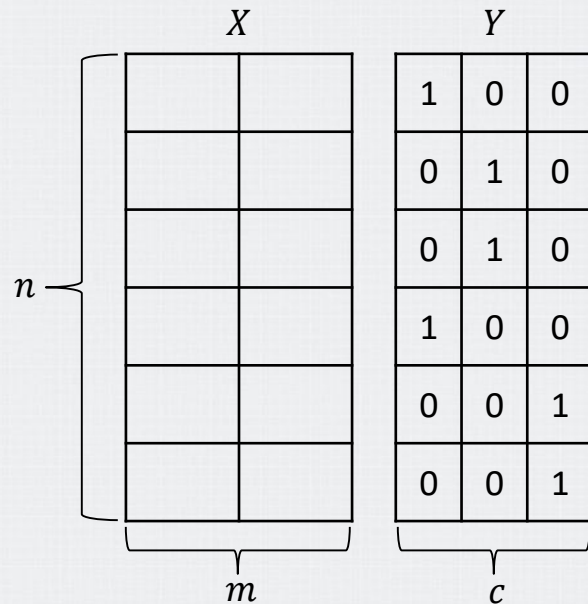
- Let $x_i \in \mathbb{R}^{1 \times m}$ be the *ith* sample (example) of X
 - We can express x_i in terms of its features $x_i = x_i^1, x_i^2 \dots x_i^m$
- Let $y_i \in \mathbb{R}^{1 \times 1}$ be the *ith* label of Y

X		Y
0.69	10.7	0
0.85	14.5	0
x_i 0.67	20.3	0 y_i
....		.
0.32	6.1	1
0.21	4.0	1

Multiclass Problems

Independent and Dependent Variables

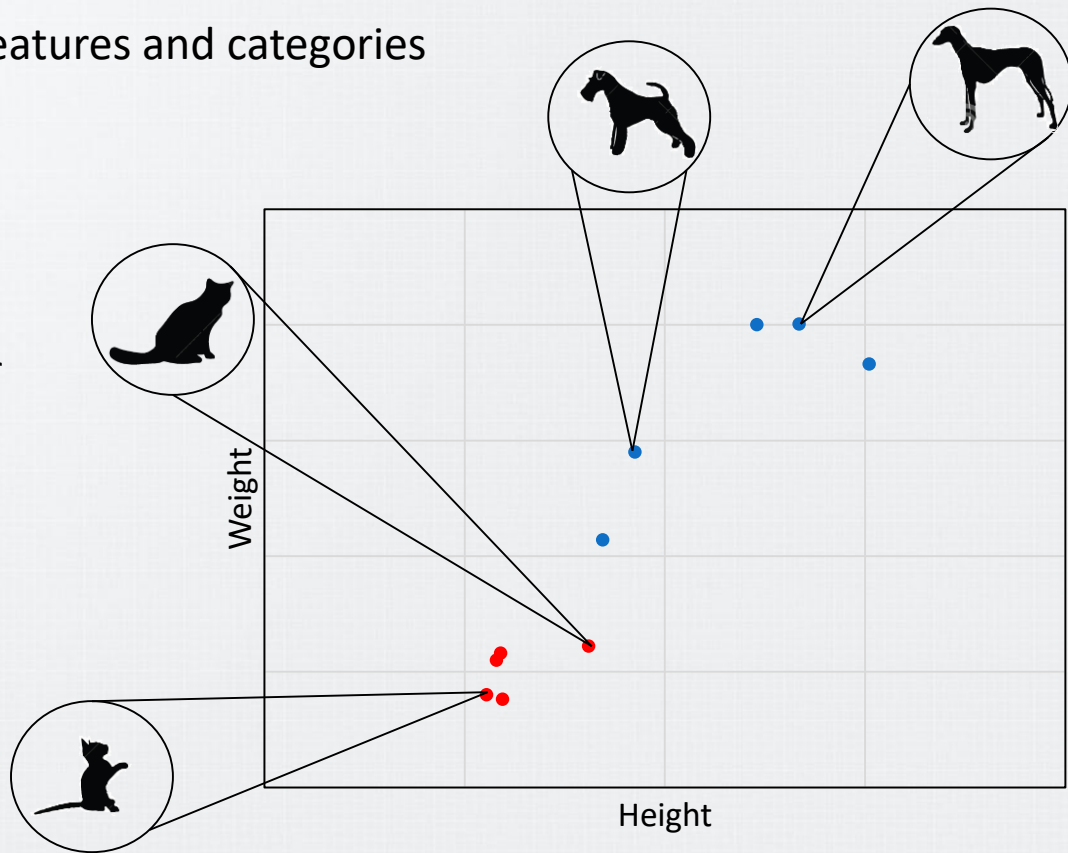
- Problems involving more than two classes
 - In this case, we cannot use 0 and 1
 - For example, dogs, cats and bears
- One-hot encoding
 - Transform the c classes into a zero-one vector
 - The cth entry equal to 1 and the rest 0
- $Y \in \mathbb{R}^{n \times c}$



Feature Space

Independent and Dependent Variables

- Projects the data using its features and categories
- Technical details
 - Each feature (m) is an axis
 - Each sample (n) is a point
 - Each category (y) is a color



Problem Definition

Independent and Dependent Variables

- Given X , the problem is to predict Y to new samples
 - Given the features, we want to predict the category to which samples belong
- Assume $\mathcal{F}(\cdot, \cdot)$ a model parameterized by a set of parameters/weights θ that receives x and outputs $\hat{y}$
 - Formally, $\hat{y} = \mathcal{F}(x, \theta)$ (or $\hat{Y} = \mathcal{F}(X, \theta)$)
- The problem is, therefore, to discover θ**

X		$\hat{Y}$ (unseen)
0.33	8.7	?
0.78	3.5	?
0.29	7.3	?
....		.
0.62	9.1	?
0.35	10.0	?

Problem Definition

Independent and Dependent Variables

- Training (or learning) phase
 - Estimate θ using X and Y (seen)
- Testing phase
 - Employ θ onto X to predict $\hat{Y}$ (unseen)
 - Unseen means new samples

X		Y (seen)
0.69	10.7	0
0.85	14.5	0
0.67		.
....	6.1	1
0.32	4.0	1

Training Phase

X		$\hat{Y}$ (unseen)
0.33	8.7	?
0.78	3.5	?
0.29		.
....	9.1	?
0.62	10.0	?

Testing Phase

Classification vs. Regression

Independent and Dependent Variables

- Classification
 - The goal is to predict a category between c possible categories
- Regression
 - The goal is to predict a real-valued target
- For both classification and regression, the problem is to discover θ
 - $\hat{Y} = \mathcal{F}(X, \theta)$

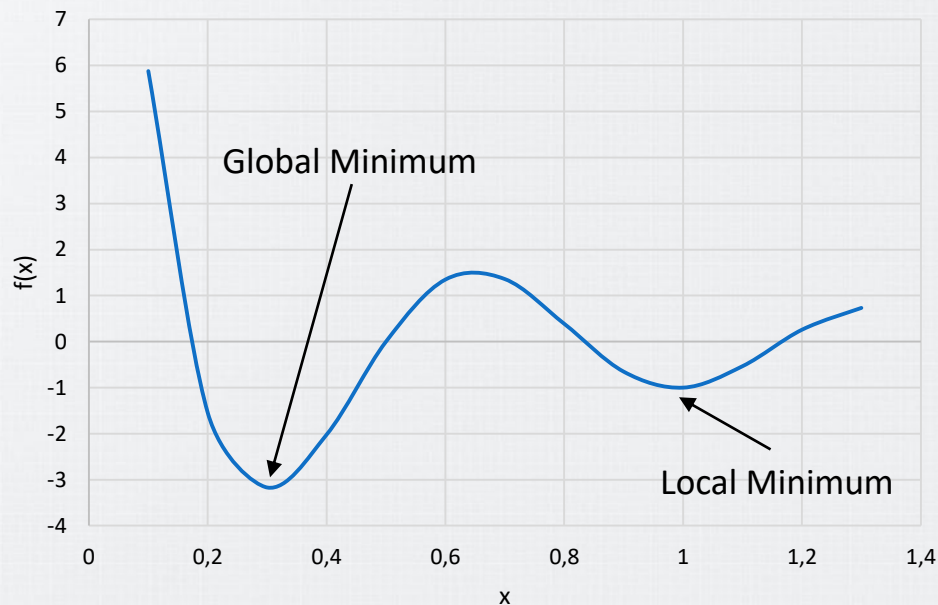
X (features)	Task	Y (target)
Weight, Height (Animal features)	Classification	Dogs, Cats, Bears
	Regression	Lifespan
Area, location, number of bedrooms (House features)	Classification	House Quality (good, bad, medium)
	Regression	House Price

Preliminaries

Functions

Preliminaries

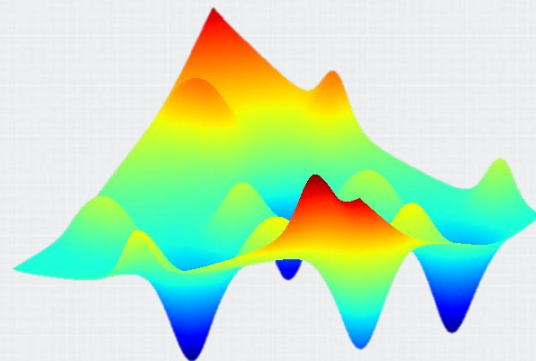
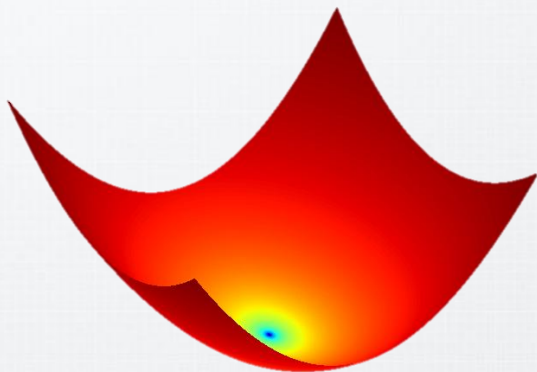
- A function is a relation that associates each element x to **a single** element $f(x)$
- Global minimum
- Local minimum



Convex and Non-Convex Problems

Preliminaries

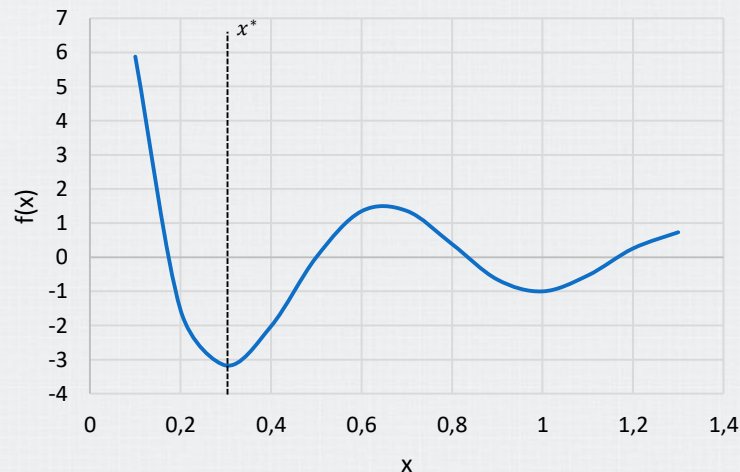
- Convex Problems: any local minimum is also a global minimum
 - It facilitates the optimization process (i.e., discovering θ)
- Non-Convex Problems: local minima are not necessarily global minima
 - Multiple optimal local solutions
 - Most practical problems belong to this category – deep learning problems



Loss Function

Preliminaries

- We name the function that we want to minimize or maximize the objective function or criterion
 - When we are minimizing a function, we may also call it the **cost function**, **loss function**, or **error function**
- We often denote the value that minimizes (or maximizes) a function with a superscript *
 - $x^* = \operatorname{argmin} f(x)$



Loss Function

Preliminaries

- Quantify the distance between the real (Y) and predicted values of the target
 - Real values: y or Y
 - Predicted values: $\mathcal{F}(x, \theta) = \hat{y}$ or $\mathcal{F}(X, \theta) = \hat{Y}$
- Loss function properties
 - Monotonicity: The better the model gets, the lower the value of the loss function
 - Differentiability: Differentiable with respect to θ

Common Losses	Short Description
Squared error	$\frac{1}{2}(\hat{y} - y)^2$
Mean Square Error	$\frac{1}{n} \sum_{i=0}^n (\hat{y}_i - y_i)^2$
Cross Entropy	$-\sum_{i=0}^c y_i \log(\hat{y}_i)$

The Role of the Derivative

Preliminaries

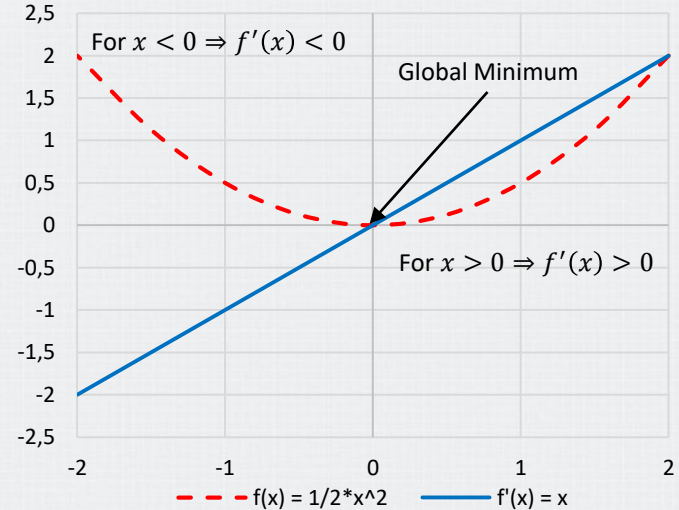
- Define a function $y = f(x)$ and $f'(x)$ the derivative of f
 - x and y are both real numbers
- The derivative $f'(x)$ gives the slope of $f(x)$ at the point x
 - It specifies how to scale a small change in the input in order to obtain the corresponding change in the output: $f(x + \epsilon) \approx f(x) + \epsilon f'(x)$

Diagram illustrating the definition of the derivative:

$$\frac{\partial f}{\partial x} = \lim_{e \rightarrow 0} \frac{f(x + e) - f(x)}{e}$$

Annotations:

- $\frac{\partial f}{\partial x}$: Quantity (w.r.t. Time interval)
- e : Change in the value of quantity (Tends to zero (becomes very small))
- $f(x + e)$: New value
- $f(x)$: Old value



The Role of the Derivative

Preliminaries

- The **gradient** generalizes the notion of derivative to the case where the derivative is with respect to a vector
- The gradient of f is the vector containing all of the partial derivatives $\nabla_x f(x)$
 - Element i of the gradient is the partial derivative of f with respect to x_i

[Hands-on](#)



Normalization

Z-Score

Normalization

- Suppose we are mapping the world using the following features
 - $x^1 \in [0, 100], x^2 \in [0, 1], \dots, x^m \in [-\infty, +\infty]$

$$\hat{y} = w_1 * x^1 + w_2 * x^2 + \dots + w_m * x^m + b$$

Diagram illustrating the mapping of feature ranges to the linear model equation:

- $[0, 100]$ maps to x^1
- $[0, 1]$ maps to x^2
- $[-\infty, +\infty]$ maps to x^m

$X \in \mathbb{R}^{n \times m}$

- Z-score normalization

- $$X \leftarrow \frac{X - \mu}{\sigma}$$

$\mu \in \mathbb{R}^{1 \times m}$
(average sample)

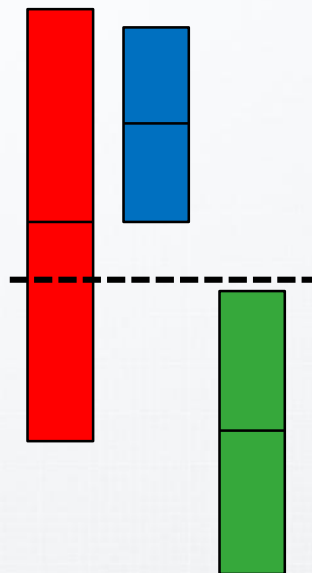
--	--	--	--

$\sigma \in \mathbb{R}^{1 \times m}$

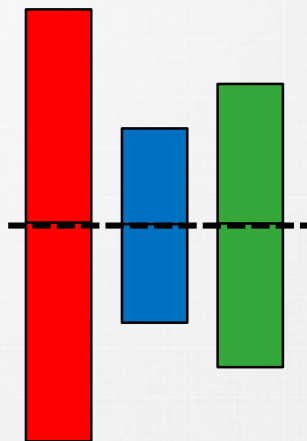
--	--	--	--

Z-Score

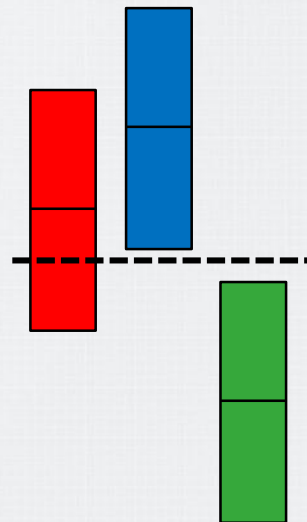
Normalization



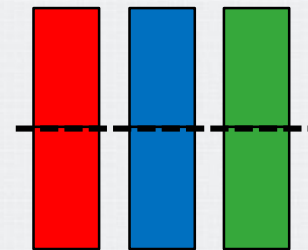
Raw data



Zero Mean



One Deviation

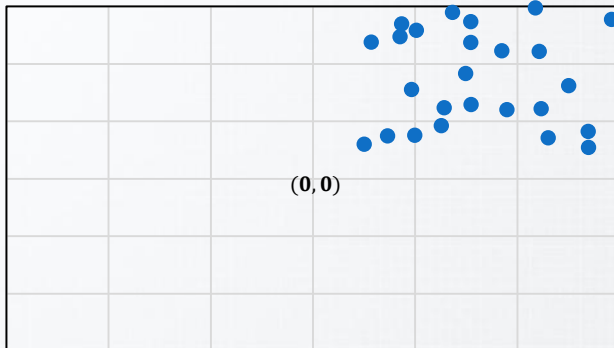


Z-Score

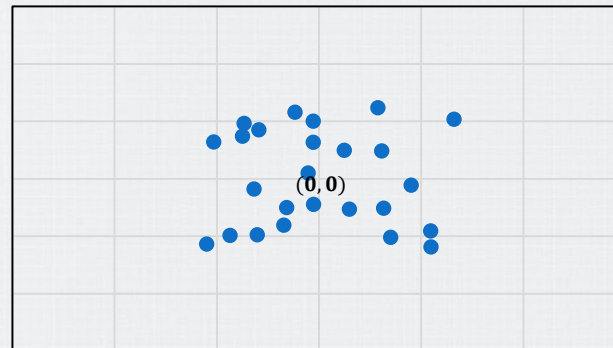
Z-Score

Normalization

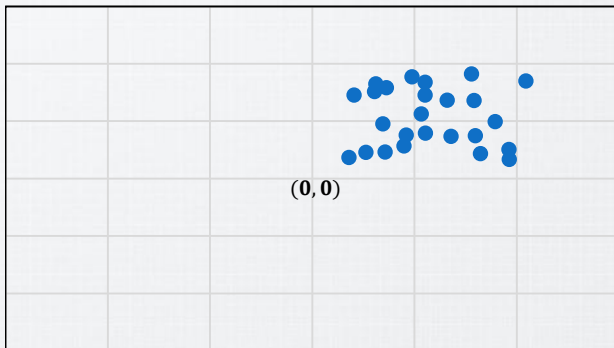
Raw Data



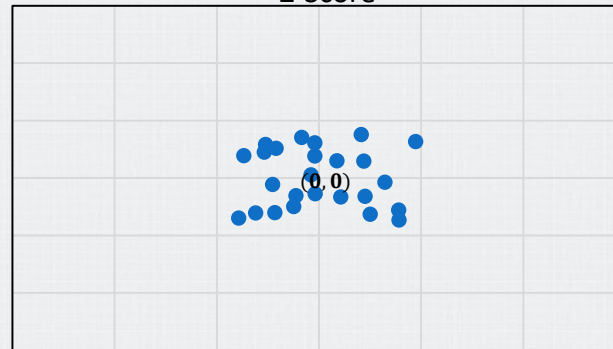
Zero Mean



One Deviation



Z-Score



Average Sample

Normalization

X



μ (average sample)



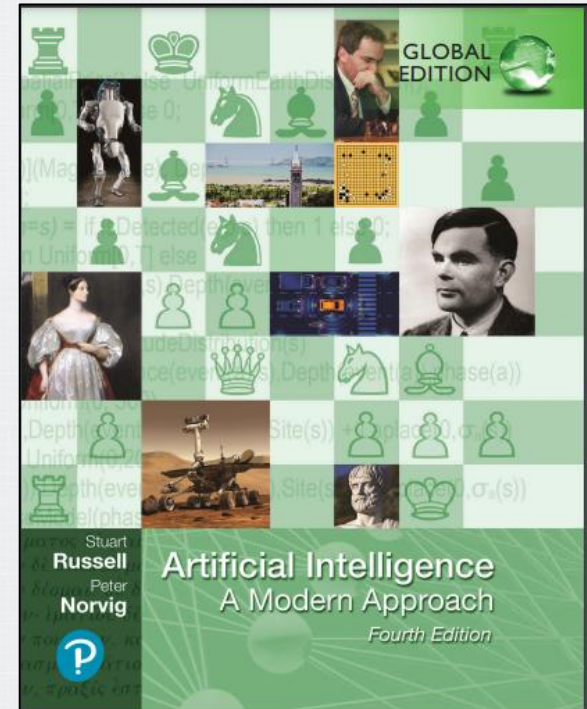
[Hands-on](#)



Bibliography

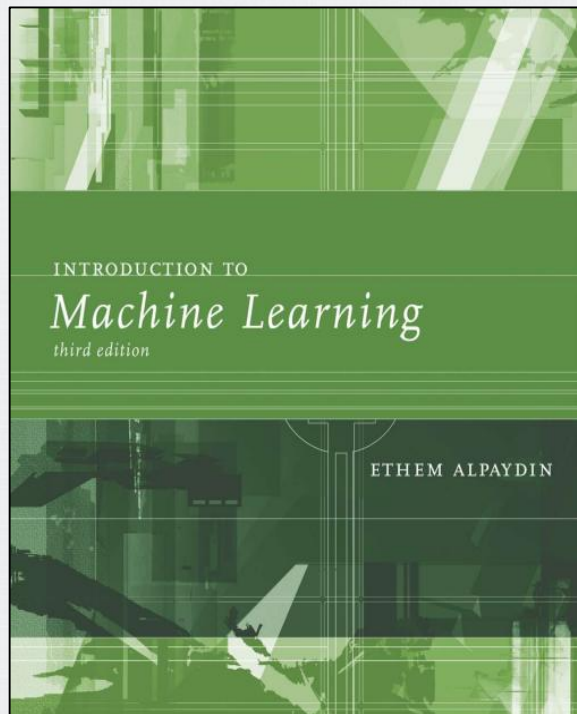
Bibliography

- Artificial Intelligence A Modern Approach



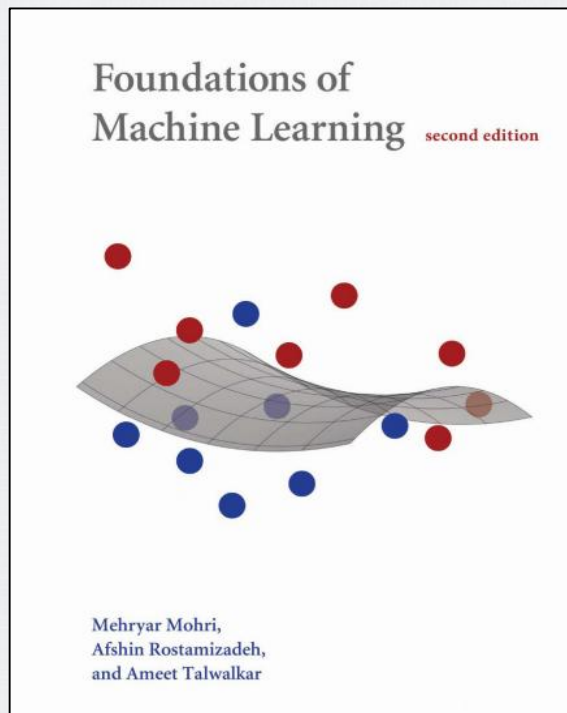
Bibliography

- INTRODUCTION TO Machine Learning



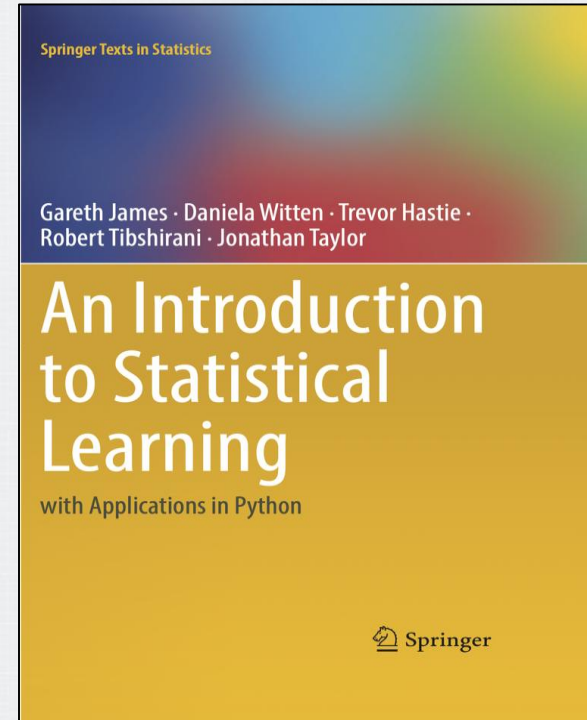
Bibliography

- Foundations of Machine Learning



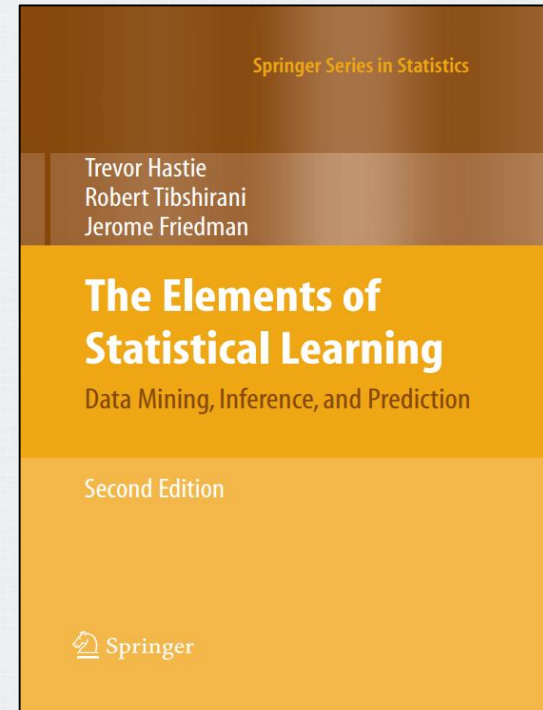
Bibliography

- An Introduction to Statistical Learning



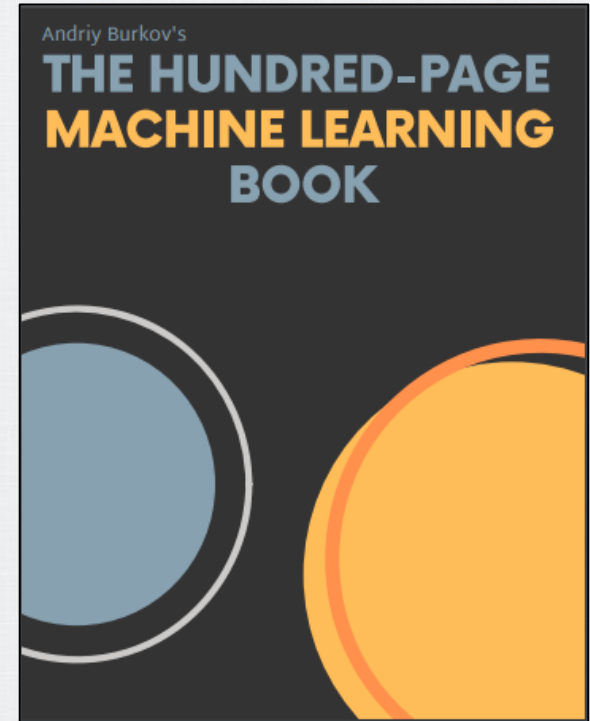
Bibliography

- The Elements of Statistical Learning



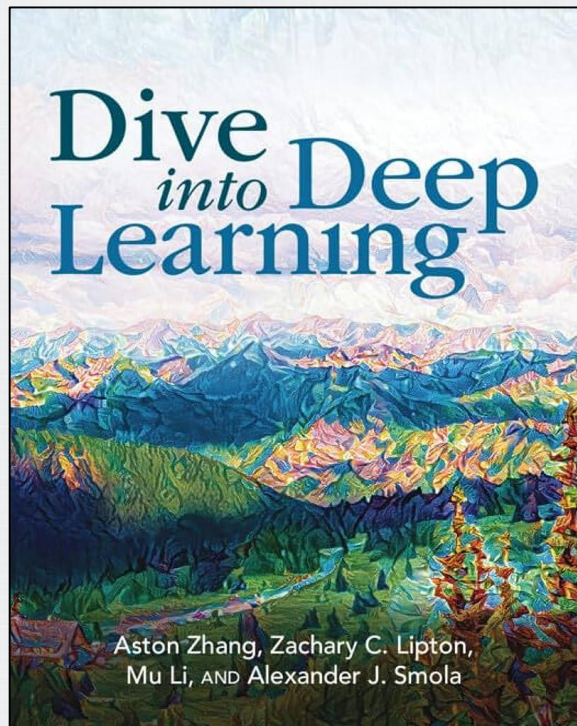
Bibliography

- The Hundred-page Machine Learning Book



Bibliography

- Dive into Deep Learning



Bibliography

- Deep Learning

